Revealing the Milky Way's Center. Image credit: NASA, JPL-Caltech, Susan Stolovy (SSC/Caltech) et al. (2019). https://science.nasa.gov/resource/revealing-the-milky-ways-center/

Hands on astronomical data sonification

Analyzing the universe through sound

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Session 4: Using and evaluating sonification tools. Galaxy classification with ViewCube

1. Evaluation/Validation frameworks
2. Common validation methods
3. Evaluation metrics: accuracy, readability, aesthetics, reproducibility, and generalization
4. Evaluation design in 5 steps
5. Exercises: Galaxy classification. The Hubble sequence. Using ViewCube for the classification of galaxies through sound
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1. Evaluation/Validation frameworks

The evaluation and validation of sonification projects usually combine perceptual testing, task-based performance measures, and real-world usability checks. The core idea is to test not just whether the sounds are interesting, but whether they help people perceive, learn, and use the data effectively.

A good framework include three main questions:

- Does the mapping make sense perceptually?
- Can users perform the intended task?
- Does the system work in practice in its target context?

The evaluation method should match the project goal, and should begin early rather than after the design is finished. A practical sonification evaluation framework can be organized into these layers:

- **Mapping validity**: Are the data-to-sound mappings interpretable, learnable, and consistent with the task? Test whether pitch, timbre, rhythm, spatialization, and other parameters communicate the intended data relations.

- **Perceptual effectiveness**: Can listeners discriminate, identify, rank, or compare sonified events reliably? Use identification, attribute ratings, discrimination, dissimilarity ratings, and sorting as standard perceptual methods.
- **Task performance**: Does sonification improve detection, tracking, estimation, pattern recognition, or decision-making compared with a baseline such as silence, visual-only display, or an alternative sonification?
- **Usability and ecology**: Does it work in the intended environment with the target users, noise conditions, cognitive load, and interaction pattern? Laboratory success may not transfer to field use, so ecological testing is often needed.
- **Longer-term adoption**: Can users learn it, remember it, and use it repeatedly without fatigue or confusion? This matters especially for monitoring and interactive systems.

Task	Typical Measures	Typical Usage
Identification	▪ Accuracy (% correct) ▪ Reaction time for correct ID	▪ Design and selection of sounds that are perceptually distinct from each other or that are inherently "meaningful"
Attribute Ratings	▪ Rating scale (e.g. 1 to 7, where 1 = 'very unpleasant' to 7 = 'very pleasant')	▪ Discovery of relationships between perceptual and acoustic properties of sounds. ▪ "Labeling" dimensions that determine similarities and differences between sounds. ▪ Input data for factor analysis and other techniques for determining "structure" among a set of sounds.
Discrimination	▪ Accuracy (% correctly compared) ▪ Errors (number and type of incorrect responses)	▪ Design and selection of sounds that are perceptually distinct from each other.
Dissimilarity Ratings	▪ Numeric estimate of similarity between pairs of sounds. ▪ Dissimilarity or proximity matrix.	▪ To determine which sounds are highly similar (possibly confusable) or distinct. ▪ Input data for cluster analysis and MDS for determining perceptual "structure" of set of sounds.
Sorting	▪ Dissimilarity or proximity matrix	▪ To determine which sounds are highly similar (possibly confusable) or distinct. ▪ Input data for cluster analysis and MDS for determining perceptual "structure" of set of sounds.

Table 1. Data collection methods, Bonebright and Flowers, p. 120.

2. Common validation methods and metrics

The most common validation approaches are experimental and comparative. In sonification research, a strong design usually compares multiple mappings or conditions and checks whether one performs significantly better than another for the pretended task.

Typical methods include:

- Controlled listening tests for discrimination, identification, and similarity judgments.
- Paired comparison or sorting tasks to see whether listeners recover the intended structure in the data.
- Task completion studies where participants use the sonification to solve a realistic problem.
- Expert review plus user testing, especially when domain experts and end users have different needs.
- Pilot testing to catch interface, timing, and instruction problems before the real study.

3. Evaluation metrics

Metrics should follow the use case and allow the evaluation of the accuracy, readability, accessibility, aesthetics, reproducibility, and generalization of the proposal. For exploratory sonification, accuracy in comparing patterns or finding outliers may matter most. For monitoring systems, detection time, missed events, and false alarms are often more important. For educational or outreach sonifications, learning rate, confidence, recall, and subjective feedback become relevant.

Goal	Metrics
Pattern recognition	Accuracy, response time, confusion rate
Monitoring	Event detection rate, false alarms, latency
Comparison tasks	Pairwise correctness, ranking agreement, dissimilarity structure
Usability	Completion time, workload, qualitative feedback
Learnability	Improvement across trials, retention
Aesthetics	Qualitative feedback
Generalization	Stability and validation across different case studies
Reproducibility	Available resources and documentation
Accessibility/Inclusion	Contribution, target users included, improvement

Table 2. Useful metrics associated with project goals.

4. Evaluation design in 5 steps

1. Define the user task and success criteria.
2. Pilot the stimuli and instructions.
3. Run a preliminary test and adjust it based on potential issues.
4. Run controlled perceptual tests. Compare against baselines or alternative mappings.
5. Test in the intended context and iterate based on findings.

5. EXERCISES

Exercise 1. La secuencia de Hubble (CAB-INTA)

This case study provides a great description of the Hubble sequence, commonly used for the morphological classification of galaxies. Read the following file to know more about the Hubble sequence and classify the galaxies proposed in its exercises. We will
https://svo.cab.inta-csic.es/wp-content/uploads/2026/05/EuroVO_HubbleSeq_spanish_jul2023.pdf

Exercise 2. Galaxy classification through sound using ViewCube

In this exercise we will generate a sonification of the H-R diagram of the Pleiades using Highchart Sonification Studio. Follow the steps below to generate your sonification.

1. Install ViewCube: https://viewcube.readthedocs.io/en/latest/intro.html
2. For usage instructions see: https://pypi.org/project/ViewCube/
3. Visualize the sample galaxy NGC2347. To activate the sonification press "H".
4. Can you classify the sample galaxy NGC2347 using the Hubble sequence? Can you do it by ear?
5. Working in groups of two or more participants, classify the morphology of the following galaxies using sound. One participant loads the galaxy and writes the type of galaxy looking at the screen and listening to the sonification, which is controlled by the other participant, who controls the mouse in a blind test (without seeing the screen). Both participants write their classification in the following table. Then roles are flipped for the next galaxy. At the end, use Simbad to check your results for each galaxy.

Only sound participant (blind test):

Galaxy	Elliptic	Spiral	Barred spiral	Other	Simbad

Participant with multimodal representation (graphics and sound):

Galaxy	Elliptic	Spiral	Barred spiral	Other	Simbad

Report your work

Use this form to deliver your exercises. Send a link to a pdf file stored in your own cloud drive. https://forms.gle/fUW9wDKubU85fKE38

Thanks for your participation!

Q&A

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Este documento hace uso de las guías educativas de Observatorio Virtual originalmente desarrolladas en el marco del proyecto EuroVO-AIDA (INFRA-20071.2.1/212104) y actualizadas en el marco del proyecto PID2020-112949GB-I00 financiado por: MCIN/AEI/10.13039/501100011033/

The study also uses data provided by the Calar Alto Legacy Integral Field Area (CALIFA) survey (https://califa.caha.es/). Based on observations collected at the Centro Astronómico Hispano Alemán (CAHA) at Calar Alto, operated jointly by the Max-Planck-Institut fűr Astronomie and the Instituto de Astrofísica de Andalucía (CSIC).